FIG. 12A -- MULTI-LEADER HEARTBEAT FAILOVER: TOPOLOGY HEARTBEAT PROTOCOL 1200

SECTION A: NORMAL-STATE MULTI-LEADER TOPOLOGY 1200

```
LEVIATHAN alpha
LEADS: P0
  Sequencer_0, WAL_0
  BatchTimer_0
REPLICATES: P1,P2 (RO)
MessageStore (GLOBAL)
```

```
LEVIATHAN beta
LEADS: P1
  Sequencer_1, WAL_1
  BatchTimer_1
REPLICATES: P0,P2 (RO)
MessageStore (GLOBAL)
```

```
LEVIATHAN gamma
LEADS: P2
  Sequencer_2, WAL_2
  BatchTimer_2
REPLICATES: P0,P1 (RO)
MessageStore (GLOBAL)
```

DETERMINISTIC FAILOVER CHAIN (pre-configured, static, never changes at runtime):

```
Partition 0:
  [alpha,beta,gamma]
alpha=PRIMARY (priority 0)
beta=backup-1 (priority 1)
gamma=backup-2 (priority 2)
```

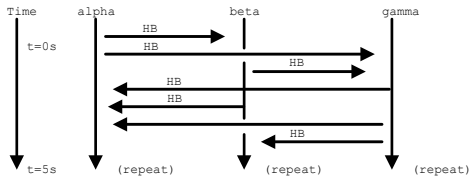
```
Partition 1:
  [beta,gamma,alpha]
beta=PRIMARY (priority 0)
gamma=backup-1 (priority 1)
alpha=backup-2 (priority 2)
```

```
Partition 2:
  [gamma,alpha,beta]
gamma=PRIMARY (priority 0)
alpha=backup-1 (priority 1)
beta=backup-2 (priority 2)
```

Rule: LOWEST INDEX = HIGHEST PRIORITY. Highest-priority LIVE node leads. No election.

SECTION B: HEARTBEAT BROADCAST PROTOCOL 1202

Heartbeat Sequence (every 5,000ms):



HB = leader:heartbeat gossip
Broadcast to ALL connected peers

LeaderHeartbeatPayload:

```
nodeId: "alpha"
partitions: [0] (led partitions)
partitionSequences: {0: "42"}
timestamp: 1709312000000
publicKeyHex?: "a1b2..." (Ed25519)
```

Leader Tracking -- leaderMap per node:

```
Example -- beta's leaderMap:
P0: leader=alpha, lastHB=t, local=N
P1: leader=beta, lastHB=t, local=Y
P2: leader=gamma, lastHB=t, local=N
```

Heartbeat interval: 5,000ms
Heartbeat timeout: 15,000ms (3 missed = dead)